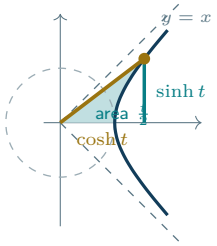
FUNCTIONS LADDER 2 OF 10

Hyperbolic functions

The point $(\cosh t, \sinh t)$ runs along a hyperbola as $(\cos t, \sin t)$ runs round a circle.

1 / THE RULE AND THE PICTURE

The point $(\cosh t, \sinh t)$ lies on $x^2 - y^2 = 1$, and t is twice the shaded area



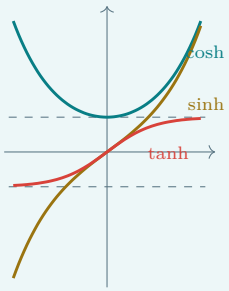
Hyperbolic cosine and sine are the even and odd parts of the exponential:

$$\cosh t = \tfrac{e^t + e^{-t}}{2}, \qquad \sinh t = \tfrac{e^t - e^{-t}}{2}, \qquad \cosh^2 t - \sinh^2 t = 1$$

The identity follows from $\frac{1}{4}\big((e^t + e^{-t})^2 - (e^t - e^{-t})^2\big) = \frac{1}{4} \cdot 4 = 1$, and the sector from the origin to the point has area $t/2$, just as the circular sector has area $\theta/2$.

2 / THE GRAPHS / THREE CURVES

Cosh is a symmetric valley, sinh an odd curve, and tanh squashes into $(-1, 1)$



$\cosh t \geq 1$ with minimum 1 at $t = 0$, $\sinh t$ passes through the origin with slope 1, and $\tanh t = \sinh t / \cosh t$ rises from -1 to 1.
For large t both $\cosh t$ and $\sinh t$ are close to $\frac{1}{2}e^t$. Unlike the circular functions none of them is periodic.

3 / VALUES AND IDENTITIES

The identities are those of trigonometry with a sign change

t	$\sinh t$	$\cosh t$	$\tanh t$
0.5	0.5211	1.1276	0.4621
1	1.1752	1.5431	0.7616
2	3.6269	3.7622	0.9640

Osborn's rule says to take a trigonometric identity, write each function as its hyperbolic counterpart, and flip the sign of any product of two sines: $\cosh(A + B) = \cosh A \cosh B + \sinh A \sinh B$, $\sinh 2t = 2 \sinh t \cosh t$, $\cosh 2t = \cosh^2 t + \sinh^2 t = 2\cosh^2 t - 1$ and $1 - \tanh^2 t = \operatorname{sech}^2 t$.
Check with $A = 1$, $B = 2$: $1.5431 \cdot 3.7622 + 1.1752 \cdot 3.6269 = 10.0677 = \cosh 3$.

4 / CALCULUS AND INVERSES

The derivatives are $\frac{d}{dt}\sinh t = \cosh t$ and $\frac{d}{dt}\cosh t = \sinh t$, with no minus sign

The inverses are logarithms, found by solving $\sinh y = x$ as a quadratic in e^y: $\operatorname{arsinh} x = \ln\big(x + \sqrt{x^2 + 1}\big)$, $\operatorname{arcosh} x = \ln\big(x + \sqrt{x^2 - 1}\big)$ for $x \geq 1$, and $\operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x}$. For example $\operatorname{arsinh} 1 = \ln(1 + \sqrt{2}) = 0.8814$ and $\operatorname{artanh}\frac{1}{2} = \frac{1}{2}\ln 3 = 0.5493$.

5 / THE HANGING CHAIN / THE CATENARY

A uniform chain hangs in the shape $y = a\cosh(x/a)$

For $a = 1$ m a chain hung from two points 2 m apart, at $x = \pm 1$, sags by $a(\cosh 1 - 1) = 0.543$ m below its ends, and its length is $2a \sinh 1 = 2.350$ m, since the arc length from 0 to x is $a\sinh(x/a)$.
A parabola looks similar near the bottom but is not the same curve, and the inverted catenary is the ideal shape for an arch.

ASIDE / THE CONNECTION

With complex arguments the two families merge: $\cos(ix) = \cosh x$ and $\sin(ix) = i \sinh x$.

LINKS TO OTHER SHEETS

The previous sheet is *Circular functions*; the next is *Small-angle approximations*; Euler's formula explains the link in *Euler's formula and the complex exponential*.

Convention. t is a real number; $\operatorname{sech} = 1/\cosh$; arsinh etc. are inverse hyperbolic functions; a is a length.